Real-time and the power budget

IoT · University · Lecture 2



In this lecture

- Budget energy with duty cycles.
- Define hard and soft real-time.
- Check schedulability.

Remember

- life = capacity / I_avg.
- The radio dominates.
- Real-time means on time.

Measure, then schedule

1. Warm-up (5 min): Where did the energy go?
2. Measure (20 min): Current.
3. Budget (15 min): Battery life.
4. Schedule (15 min): Three tasks.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. How far was the datasheet from your measurement?
2. Is your system hard or soft real-time?
3. What would you send less often?

Exit ticket

- Compute the battery life of a node at 0.25 mA average on 1,000 mAh.